

Retail Revenue Intelligence

Unlocking Sales Performance Through Data-Driven Insights

TEAM MEMBERS

Divy Kumar Jain (Lead) • Tanisha Dhiman • Kartikey Gupta • Disha Khanka • Yash Raj

SECTOR

Retail / Consumer Goods

TEAM ID

Section D — Group 5

MENTOR

Satyaki Das

DATE

29 April 2026

Context & Problem Statement

Sector Context

Multi-category retail spanning Beverages, Butchers, Computers, Food, Furniture, Milk Products & Patisserie — operating across Online & In-Store channels.

Core Business Question

What truly drives revenue across product categories, channels, customers, and promotions — and which levers should be prioritised to maximise growth?

Stakeholder

Retail business managers, category heads & operations directors making pricing, inventory, promotion and channel investment decisions.

Objective

Transform 12,575 raw POS transactions into a production-grade Revenue Intelligence system with statistical insights and interactive Tableau dashboards.

Data Engineering

12,575

Raw Records

11 Fields

Columns

3 Years

Jan 2022 – Dec 2024

11,971

Clean Records

ETL Pipeline — 8 Key Steps

1. Column standardisation & type conversion
2. Duplicate removal (0 found across 12,575 IDs)
3. Missing value imputation (price × qty logic)
4. Discount nulls → False; item nulls → 'Unknown'
5. Invalid row removal (non-positive values)
6. Integrity check: price × qty = total (100% pass)
7. Categorical standardisation (Title-Case)
8. Feature engineering: year, month, year_month

Source: retail_store_sales.csv | 25 Unique Customers | 8 Categories | 200 Unique Items | Output: cleaned.csv (11,971 × 11)

KPI Framework

Total Revenue

Rs. 15,52,071

Top-line health indicator

Avg Order Value (AOV)

Rs. 129.65

Revenue efficiency per transaction

Revenue by Category

Butchers Highest

Identify winning departments

Revenue by Payment

Cash Rs. 5.38L

Payment preference mapping

Customer Total Spend

Rs. 53K–72K

High vs low value customers

Total Transactions

11,971

Volume & demand intensity

Total Qty Sold

66,268 units

Throughput & inventory planning

Revenue by Location

Online Rs. 7.91L

Omni-channel balance tracking

Monthly Revenue Trend

Rs. 33K–53K/mo

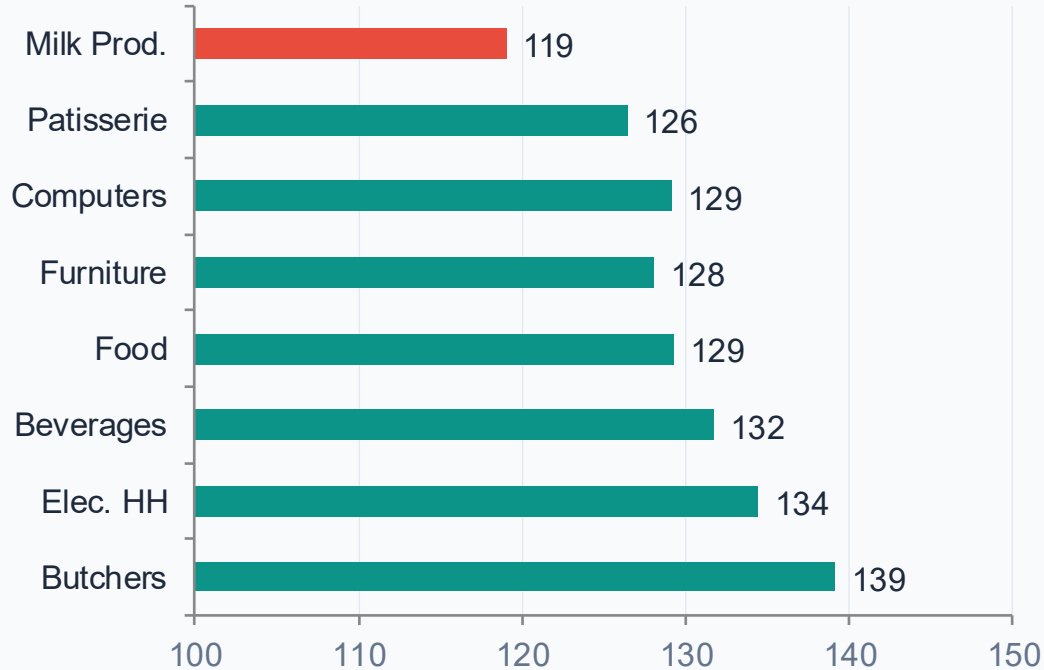
Seasonality & growth trajectory

Discount Impact

Rs. 130 vs Rs. 129

Discount programme ROI

Key EDA Insights



Butchers leads at AOV Rs. 139 — Rs. 20 above Milk Products



Discounts show NO impact: $p=0.49$, Cohen's $d=0.01$



Online vs In-Store nearly equal: Rs. 7.91L vs Rs. 7.61L



Qty ($r=0.71$) beats Price ($r=0.63$) as revenue driver



No seasonal pattern — ANOVA $p=0.35$ across 12 months



Revenue right-skewed: median Rs. 109 vs mean Rs. 130

Advanced Analysis

Statistical Hypothesis Tests (95% Confidence Level, $\alpha = 0.05$)

Test	Hypothesis	Result	Finding
Welch's t-test	Discounts lift basket size?	$p = 0.493 \times$	Discounts have zero impact
One-Way ANOVA	Categories differ in AOV?	$p < 0.001 \checkmark$	Butchers leads; Milk lags
One-Way ANOVA	Payment method affects spend?	$p = 0.497 \times$	No payment advantage
One-Way ANOVA	Online vs In-Store spend?	$p = 0.368 \times$	Genuine omni-channel balance
One-Way ANOVA	Monthly seasonality?	$p = 0.346 \times$	No seasonal pattern detected

Pearson Correlation — Revenue Drivers

Quantity → Total Spent $r = 0.712$ (Strong Positive)

Price Per Unit → Total Spent $r = 0.631$ (Moderate-Strong)

→ Selling more units per transaction outperforms price increases as a revenue strategy

Dashboard Overview

01 Executive Overview

High-level KPIs: Total Revenue, Transactions, AOV & Discount Rate. Monthly revenue trend line chart. Category revenue rankings. All interactive.

02 Customer Analysis

Per-customer total spend & transaction frequency. Identifies CUST_12 as top spender (Rs. 71.8K). Highlights low-engagement customers for retention.

03 Sales & Category

Category-level revenue bars, payment method distribution (Cash/CC/Wallet), online vs in-store splits, and discount impact visualisation.

Interactive Filters:

▸ Category (All 8 product groups) ▸ Year (2022 / 2023 / 2024) ▸ Location (Online / In-Store)

Top Decision Insights

1

Discounts Are Margin-Destructive

Welch's t-test $p=0.49$, Cohen's $d=0.01$. The Rs. 1.26 AOV difference is neither statistically nor practically significant. Blanket discounts silently erode gross margin.

2

Quantity is the #1 Revenue Lever

Pearson $r=0.71$ — stronger than price ($r=0.63$). Adding one unit per transaction has more impact than raising prices. Multi-buy tactics are the highest ROI play.

3

Butchers & Milk Products: Two Tiers

Non-overlapping 95% CIs confirm a real performance gap. Butchers (Rs. 134–144 CI) vs Milk Products (Rs. 114–124 CI). Intervention is data-backed, not intuition.

4

Omni-Channel Balance is a Strength

Rs. 7.91L Online vs Rs. 7.61L In-Store — ANOVA $p=0.37$. The near-perfect balance is a competitive advantage. Maintain proportional investment; don't disrupt it.

Recommendations

HIGH

🕒 4–6 wks

Restructure Discount Programme

Replace blanket % discounts with conditional 'Buy 3 Get 1 Free' promotions on Milk Products & Patisserie. Est. uplift: +Rs. 46,800/year.

HIGH

🕒 2–8 wks

Quantity-Driven Upselling

Multi-buy bundles, 'frequently bought together' prompts online, and in-store upsell scripts. 10% qty lift → +Rs. 1.55L/year.

MED

🕒 6–10 wks

Milk Products Growth Plan

Premium sub-line (artisanal/organic), cross-category bundles with Butchers/Beverages, dedicated promo space. Est. +Rs. 15,130/year.

MED

🕒 4–6 wks

Tiered Loyalty Programme

Silver / Gold / Platinum tiers by cumulative spend. Rewards: early access, free delivery, exclusive pricing. Est. 5% lift → +Rs. 77,600/year.

LOW

🕒 Ongoing

Retailer-Driven Demand Events

3 quarterly flash sales targeting sub-Rs. 40K months. 15% lift per event → +Rs. 17,500/year. Manufactures peaks where none exist naturally.

Impact & Feasibility

Upselling
(Qty +10%)

+Rs. 1.55L

per year

IMMEDIATE

Feasibility: High

Discount
Restructure

+Rs. 46.8K

per year

IMMEDIATE

Feasibility: High

Milk Products
Growth

+Rs. 15.1K

per year

SHORT-TERM

Feasibility: Medium

Loyalty
Programme

+Rs. 77.6K

per year

MED-TERM

Feasibility: Medium

Combined Annual Revenue Uplift Potential:

~Rs. 2.95 Lakhs / year

Limitations



Small Customer Base (n=25)

Per-customer insights carry low statistical power. Results are directional — not generalisable to larger retail populations without validation.



Anonymised Item Codes

Item_6_FOOD style codes prevent product-name analysis. All analytics are conducted at the 8-category level only.



No Cost / Margin Data

Revenue-only dataset. Cannot compute gross margins or true promotion ROI. Analysis is confined to top-line revenue metrics.



No Demographic Data

Customer age, gender, income, and geography are unavailable. Segmentation is limited to behavioural dimensions (spend, frequency).



33.4% Missing Discount Data

All nulls imputed as False (no discount). May understate true discount utilisation. Sensitivity analysis is recommended future work.



3-Year Static Snapshot

Dataset does not capture real-time dynamics or external shocks. All findings are descriptive of 2022–2024 only; causal claims are avoided.

Next Steps & Future Extensions

github.com/kartikey0104/
SectionD_Group5_
RetailRevenue
Intelligence

- 01 Predictive Forecasting** — ARIMA / Prophet for 30–90 day category demand, enabling proactive procurement.
- 02 Profitability Analysis** — Integrate COGS data to shift optimisation from revenue to gross margin.
- 03 Market Basket Analysis** — Apriori algorithm to identify co-purchase patterns for bundle pricing.
- 04 RFM Customer Segmentation** — Recency-Frequency-Monetary model on a larger dataset for hyper-targeted marketing.
- 05 Real-Time Dashboarding** — Connect Tableau to Snowflake/BigQuery with daily automated ETL for live reporting.

**3
Dashboards
Published**

Section D
Group 5
2026

Raw data → Statistical truth → Actionable strategy. Revenue intelligence delivered.